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Twilight Wolfpack Analytics – Prediction Interval

Overview

In addition to point spread predictions, we constructed prediction intervals for the 78 ACC men's basketball games to quantify uncertainty around each predicted spread. These intervals represent plausible ranges for the final **point differential (home score minus away score)** rather than raw points scored.

Method

Prediction intervals were generated using the same baseline linear regression model used for point spread prediction. The model includes categorical effects for home and away teams, allowing it to estimate relative team strength and home-court advantage. Standard regression-based prediction intervals were computed using the model's estimated residual variance.

For each game, the fitted model produces:

- A point prediction for the spread
- A lower and upper bound defining the prediction interval

This approach incorporates both model uncertainty and game-to-game variability.

Interpretation

Prediction intervals may include negative or positive values. A negative lower bound indicates the possibility that the away team could win, while a positive upper bound reflects the potential margin of a home-team victory. Interval bounds are reported as continuous values, consistent with the underlying regression model and competition guidelines.

Coverage Considerations

Intervals were constructed conservatively to prioritize coverage. Historical model residuals were used to calibrate uncertainty, with the goal of achieving at least **70% coverage** on past data, as recommended by the competition rules. While narrower intervals are more precise, ensuring adequate coverage was treated as the primary objective.